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Default Risk, Productivity, and the Environment: Theory and Evidence from U.S. Manufacturing

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Abstract
This paper employs a general equilibrium model with heterogeneous firms to analyze the effect of default risk on production-generated pollution emissions. The model analytically divides the effect of default risk into three distinct effects: the market-size, technology-upgrading, and selection effect. Conceptually, an increase in default risk raises equilibrium borrowing costs, thereby precluding investment in a technology upgrade among a subset of firms (technology-upgrading effect). As a consequence, the economy consists of more numerous (market-size effect) but less productive and more pollution-intensive firms (selection effect). Because the effects are confounding in nature, the effect of default risk on aggregate pollution emissions and emissions intensity is an empirical question. To answer this question, this paper estimates the model's key parameters using a unique dataset with establishment-level credit scores and a composite measure of pollution emissions for a panel of manufacturing firms in the United States. Using a two-step procedure where default risk is estimated in the first stage, the results indicate that the estimated elasticity of emissions intensity and productivity with respect to default risk is 0.89 and − 0.16, respectively. Next, I use the theoretical model to leverage the coefficient estimates to estimate the effect of economy-wide default risk on aggregate pollution emissions, demonstrating that default risk increases aggregate emissions and emissions intensity, primarily as a consequence of the technology-upgrading effect. Finally, this paper demonstrates that historical changes in economy-wide default risk can generate economically significant changes in pollution emissions.

Keywords Default risk · Pollution emissions · Firm heterogeneity · General equilibrium

JEL Classification D50 · L60 · Q50

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1 Introduction

Firms undertake investment decisions in the context of uncertainty with respect to remaining in the market, and banks extend credit in the context of uncertainty with respect to repayment (default risk), which bears on firm borrowing costs and in turn technology choices.[1] Technology choices include investment in technologies that reduce variable input requirements, including fuel and material requirements in production, as well as investment in pollution abatement technologies. Moreover, economy-wide default risk affects exit and entry decisions, which in the context of firm heterogeneity, bears on the number of firms in the market and their average productivity and emissions intensity. Because pollution emissions are dependent on technology decisions and the number and composition of firms in the market, default risk is inextricably linked to environmental performance. Consequently, because default risk varies systematically with macroeconomic conditions, pollution emissions are generally linked with the state of the macroeconomy, consistent with empirical studies linking recessions to increases in emissions intensity (Heutel 2012).

The contribution of this paper is threefold. First, I extend the general equilibrium model developed by Andersen (2016) to analyze the effect of default risk on pollution emissions. The model divides the impact into three distinct effects: (1) the market-size, (2) technology-upgrading, and (3) the selection effect. Second, this paper estimates the model's key parameters, demonstrating that an increase in firm-level default risk increases emissions intensity and decreases productivity among manufacturing firms in the United States. Third, this paper uses the theoretical model to leverage the parameter estimates in order to numerically quantify the effect of an increase in the economy-wide default risk on aggregate pollution emissions, and to numerically decompose the disaggregate effects as described above.

To analyze the effect of default risk on aggregate pollution emissions, this paper employs a general equilibrium model with heterogeneous firms based on the Dixit and Stiglitz (1977) model of monopolistic competition. Firm heterogeneity accounts for the wide dispersion of productivities and emissions intensities across firms. More importantly, firm heterogeneity provides a basis to analyze the effect of default risk on entry and exit decisions, as well as technology-upgrading decisions. The model demonstrates that default risk, by raising equilibrium borrowing costs, precludes investment in a technology upgrade for a subset of firms that would otherwise find the investment profitable. Moreover, in the long-run, default risk leads to an economy consisting of less productive and more pollution-intensive firms.

What are the repercussions of default risk on the environment? For the purpose of exposition, consider a reduction in default risk. Reducing default risk widens the subset of firms investing in the technology upgrade, thereby increasing productivity and output among impacted firms. Consequently, pollution emissions increase through the firm-level "scale effect." On the other hand, the technology upgrade decreases emissions intensity through the firm-level "technique effect" whenever productivity and emissions intensity are inversely related, as evidenced by several empirical studies (Cole et al. 2005, 2008; Shapiro

[1] Default risk is defined as the probability that a firm (1) exits the market (goes out of business) and (2) does not repay borrowed funds in the case that the firm borrowed (or more generally the firm's creditors are repaid a lesser amount than they otherwise would receive). This paper does not consider default risk arising from other sources, such as asymmetric information (e.g., Stiglitz and Weiss 1981) and contractual incompleteness (e.g., Hart and Moore 1994).

and Walker 2015), or the technology upgrade uses a cleaner production process or mix of inputs. In sum, reducing default risk increases emissions through the scale effect and decreases emissions through the technique effect, and the net effect depends on the relative magnitude of the two opposing partial-equilibrium effects.

Default risk is also inextricably linked to pollution emissions at the general-equilibrium level through entry and exit decisions. In particular, reducing default risk increases the scale of production among impacted firms, thereby crowding out firms that are less productive in general and more pollution-intensive in particular. Consequently, reducing default risk results in an economy consisting of less numerous firms through the "market-size effect," but firms that are, on average, more productive and less-pollution intensive through the "selection effect." Similar to the partial-equilibrium effect, the general-equilibrium effect consists of confounding effects as reducing default risk reduces emissions through the market-size effect, while it increases average output and decreases average emissions intensity through the selection effect.

The empirical analysis investigates the effect of default risk on pollution emissions and productivity for a panel of manufacturing facilities, using a dataset that combines the Environmental Protection Agency's Risk-Screening Environmental Indicators and the National Establishment of Time Series. To this end, I employ a two-step procedure to estimate the elasticity of emissions intensity and productivity with respect to default risk. In the first stage, measures of creditworthiness from Dunn and Bradstreet, among other covariates, are used to predict firm death (going out of business). In the second stage, the predictions are employed as measures of default risk (probability of firm death) using establishment-level measures of emissions intensity and productivity as dependent variables. This paper also employs a two-step bootstrap procedure to account for the presence of imputed regressors to generate consistent standard errors. The results indicate that the estimated elasticity of emissions intensity and productivity with respect to default risk are 0.89 and − 0.16, respectively. Finally, this paper exploits variation in the magnitude of default risk shocks across industries in an alternative difference-in-differences approach. The Wald estimator generated from this alternative approach yields coefficient estimates that are similar to the baseline results.

Second, this paper uses the theoretical model to leverage the parameter estimates, coupled with parameters estimated by previous studies, to numerically quantify the effect of economy-wide default risk on aggregate emissions.[2] The numerical model divides the effect of default risk according to underlying disaggregate effects: the (1) market-size, (2) technology-upgrading, and (3) selection effect. The numerical model results indicate that aggregate pollution emissions are increasing in default risk due the market-size effect, but decreasing in default risk due to the technology-upgrading and selection effects. Because the market-size effect is larger in magnitude than the technology-upgrading and selection effects, aggregate pollution emissions are increasing in default risk. That is, default risk shocks, such as those generated during recessions, would lead to an increase in aggregate pollution emissions (all else constant). The numerical model also divides the effect of default risk according to aggregate technique (emissions intensity) and scale (output) effects, where the results indicate that pollution emissions are increasing due the technique effect and decreasing due the scale effect, but the technique effect is significantly larger in magnitude. Using historical default risk shocks, this paper sheds light on the magnitude of

[2] This paper also uses a two-step bootstrap procedure to estimate consistent standard errors for all of the estimated numerical model results.

the numerical model results, demonstrating that changes in economy-wide default risk can generate economically significant changes in pollution emissions and emissions intensity in particular.

Because economy-wide default risk is shaped by a wide-range of factors, including laws and institutions, fiscal and monetary policy, and regulations, this paper has implications with respect to environmental policy. For example, empirical evidence documents that environmental regulations increase default risk, as evidenced by an increase in firms exiting the market (Greenstone et al. 2012), implying that reductions in pollution from regulations might be offset by greater default risk. Consequently, optimal environmental policy might include provisions that mitigate the impact on default risk. To give a concrete example, while greater severity of punishments for non-compliance has the advantage of reducing enforcement cost for a given expected penalty, an environmental agency might prefer policy instruments with relatively less severe punishments (and relatively more monitoring) in order to mitigate the increase in default risk.[3] More generally, because default risk operates through borrowing costs and technology adoption, this paper also sheds light on the effect of various factors bearing on borrowing costs, such as imperfect information and contract enforcement, as well as public policies aimed at reducing borrowing costs (e.g., loan guarantees to small business).

This paper is related to the literature exploring the relationship between firm environmental performance and broadly-defined financial performance. Financial performance is typically defined as a measure of profitability, and the literature generally focuses on narrowly defined industries. Gray and Deily (1996) and Shadbegian and Gray (2005) find a negative relationship between profitability and compliance with environmental standards, whereas Maynard and Shortle (2001) find that more profitable and less leveraged firms are more likely to invest in a clean technology upgrade. Using a panel of Czech manufacturing firms, Earnhart and Lizal (2006, 2010) report that profits are positively associated with air pollution emissions, whereas value added is negatively associated with air pollution emissions, among Czech firms. Earnhart and Segerson (2012) contribute to the literature by pointing out that the correlation, if one exists, between financial performance and environmental performance might be the consequence of correlation between profits and unobservable factors, such as liquidity and solvency risk. Earnhart and Segerson (2012) propose a "crime and punishment" (Becker 1968) model of environmental compliance to analyze the role of liquidity and solvency risks in compliance with an environmental standard, demonstrating that the effects are confounding in nature, and that financial status does not in general play an important role in industrial wastewater discharges of chemical manufacturing facilities.[4] Using credit scores as a proxy for credit constraints, Andersen (2017) documents empirical evidence that credit constraints increase plant-level pollution emissions as a consequence of bearing on capital-investment decisions. Finally, at the country-level, Andersen (2016) uses credit market reforms as proxies for credit constraints, and documents that reducing credit constraints improves aggregate air-pollution concentrations.

This paper also contributes to the recent literature investigating the role of macroeconomic business cycles in environmental policy design (Fischer and Springborn 2011;

[3] The expected penalty is equal to the probability of punishment for non-compliance multiplied by the penalty, whereas the enforcement cost is the cost of monitoring polluters. More severe punishments might increase default risk as a consequence of triggering insolvency among liquidity constrained firms.

[4] Evans and Gilpatric (2015) develop a related model to analyze the role of liability and profitability-related risk in precautionary behavior.

Heutel 2012; Fischer and Heutel 2013). Because pollution emissions are the product of aggregate output and emissions intensity, an important policy question is whether emissions intensity changes over the business cycle. Heutel (2012) documents that the elasticity of emissions (carbon dioxide) with respect to output is positive but inelastic, implying that productivity shocks generate changes in pollution emissions that are smaller in magnitude than changes in output because emissions intensity changes over business cycles (i.e., emissions intensity increases during negative productivity shocks). Relatedly, Bae (2015) finds that plants' pollution-reduction efforts diminish during economic downturns. The question of what the underlying mechanisms are and how they operate, however, remains open.[5] The insights of this paper provide a potential explanation for the patterns of emissions intensity over the business cycles. That is, because productivity shocks are inextricably linked to changes in default risk, recessions would reduce technology upgrading, and lead to an economy consisting of less productive and more pollution-intensive firms, as a consequence of increasing economy-wide default risk.

The remainder of this paper is organized as follows. Section 2 presents the theoretical model. Section 3 discusses the data, estimation strategy, results, and numerical model simulations. Section 4 concludes.

2 Model

2.1 Overview

This section presents a general equilibrium model with heterogeneous firms based on the Dixit and Stiglitz (1977) model of monopolistic competition.[6] Time is discrete with an infinite horizon, and in each time period, a pool of initially identical potential entrants face a dynamic forward looking entry decision, where paying a market entry cost confers a draw of a random productivity parameter. Firms drawing a productivity parameter conferring present value profits in excess of a fixed production cost will take up production, while firms with lower productivity draws will immediately exit. Firms not exiting will remain in the market in subsequent periods but face an exogenous default risk; that is, probability that a random shock will force them to exit. There exists a discrete technology upgrade that is financed from external borrowing, and therefore includes an endogenous borrowing cost. With a given technology, firms produce a final good using a variable composite input, and generate pollution emissions as a byproduct of production, which depends on the technology employed.

[5] Bae (2015) posits that the association between economic downturns and pollution-reduction efforts might be due to weakened community pressure during recessions.

[6] The Dixit and Stiglitz (1977) model has been widely adopted in various economic fields, such as international trade (e.g., Melitz, 2003). Because the theoretical predictions of Melitz (2003) are consistent with empirical regularities in manufacturing sectors, and because many pollution-intensive sectors have become differentiated-good industries (such as chemicals and metals industries) (Konishi and Tarui 2015), several papers have adapted the Melitz (2003) framework to account for pollution emissions (Kreickemeier and Richter 2013; Konishi and Tarui 2015; Shapiro and Walker 2015; Andersen 2016, among others). This model extends the basic framework developed by Andersen (2016) to investigate the role of default risk in aggregate emissions. Besides focusing on default risk instead of credit constraints, the aim of the present analysis is to develop a model that can be taken to the data, wherein the micro-based structural parameters can be identified and leveraged to determine aggregate effects.

2.2 Consumers

Utility of a representative consumer exhibits constant elasticity of substitution (CES) with elasticity of substitution $\sigma > 1$

$$U = \left[\int_{v \in V} q(v)^{\frac{\sigma-1}{\sigma}} dv \right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where $q(v)$ is demand for variety v, and V is the set of available varieties.[7] Expenditure of the representative consumer is denoted Y and the price of variety v is denoted $p(v)$. Utility maximization, subject to the budget constraint $\int_{v \in V} p(v)q(v)dv = Y$, yields the isoelastic demand function for variety v

$$q(v) = YP^{\sigma-1}p(v)^{-\sigma} \tag{2}$$

where P is the CES price index.

2.3 Producers

Conditional on entry, there is a continuum of firms, each producing a unique variety (the entry decision is described in Sect. 2.3.3). Production is linear in a composite input (labeled "labor" for convenience), which includes human capital, physical capital, fuel, materials, and so on. That is, $q = \phi l$, where l is labor and ϕ is the firm's productivity parameter. The input market is perfectly competitive, with an equilibrium wage rate w, implying that marginal production costs are w/ϕ. Following Bustos (2011), the technology upgrade increases productivity by a fixed scalar parameter δ, implying that effective productivity is $\phi(1 + \delta)$ after the upgrade. That upgrading increases productivity by a constant percentage retains tractability of the model and is consistent with empirical evidence that more productive firms upgrade.[8] Consequently, marginal production costs are $w/[\phi(1 + \delta)]$ among upgrading firms. As in Bustos (2011), the technology upgrade can be interpreted, in general, to represent firm actions to reduce the marginal cost of production or to increase the quality of the product.

Conditional on entry, firms produce using either a "baseline" (non-upgraded) or upgraded technology. For notation, variables associated with the baseline technology are denoted with the superscript b, while variables associated with the upgraded technology are denoted with the superscript u. Profit maximization implies that firms set prices equal to a constant markup over marginal cost

[7] Because the model abstracts from endogenous pollution policy responses, I do not consider disutility associated with environmental damage. However, incorporating environmental damage using a weakly-separable utility function (e.g., $W(Z, U)$ where U is given by Eq. (1) and Z is aggregate pollution) would not change the analysis as the marginal rate of substitution (and thus the elasticity of substitution) of two varieties would be independent of pollution.

[8] To be precise, more productive firms will invest in the technology upgrade whenever the additional revenue generated from the upgrade is increasing in productivity, as is the case when upgrading increases productivity by a constant percentage (demonstrated below). Under empirically-relevant parameters, this condition will also hold when upgrading is additive ($\phi + \delta$), but the analysis is far less tractable and does not result in closed-form solutions.

$$p^b(v) = \frac{w/\phi}{\rho} \quad \text{and} \quad p^u(v) = \frac{w/[\phi(1+\delta)]}{\rho} \tag{3}$$

where $\rho \equiv (\sigma - 1)/\sigma$. Using the derived isoelastic demand function implies that equilibrium output and revenue for firms not investing in the upgrade are given by

$$q^b(\phi) = YP^{\sigma-1}\left(\frac{w}{\phi\rho}\right)^{-\sigma} \quad \text{and} \quad r^b(\phi) = YP^{\sigma-1}\left(\frac{w}{\phi\rho}\right)^{1-\sigma} \tag{4}$$

Moreover, firms adopting the technology upgrade set lower prices, produce more, and generate more revenue. That is,

$$\frac{q^u(\phi)}{q^b(\phi)} = (1+\delta)^{\sigma} > 1 \quad \text{and} \quad \frac{r^u(\phi)}{r^b(\phi)} = (1+\delta)^{\sigma-1} > 1 \tag{5}$$

For both non-upgrading and upgrading firms, pollution is emitted as a joint output of production, and is equal to emissions intensity (emissions per unit of output) times output. Similar to Andersen (2016), I assume that emissions intensities are given by

$$e^b(\phi) = \frac{1}{\phi^{\alpha}} \quad \text{and} \quad e^u(\phi) = \frac{\gamma}{(\phi(1+\delta))^{\alpha}} \tag{6}$$

Expression (6) reflects the net effect of upgrading on emissions intensity, where upgrading generates both indirect effects (via productivity) and direct effects on emissions intensity.[9] In particular, emissions intensity changes monotonically with firm productivity, with the technology parameter α determining the extent to which this is the case. The parameter γ reflects the difference in emissions intensity associated with the upgrade that are not due to differences in productivity.[10] The following lemma highlights the necessary and sufficient condition such that the technology upgrade reduces emissions intensity.

Lemma 1 *The technology upgrade reduces emissions intensity* ($e^b \geq e^u$) *if and only* $\gamma^{1/\alpha} \leq 1 + \delta$.

Proof Follows from expression (6). □

For simplicity and clarity, it is expedient to distinguish between output and emissions associated with the endowed baseline technology and "additional" output and emissions associated with the technology upgrade, which I denote using the superscript a. For example, additional output is given by $q^a = q^u - q^b$.

It follows that additional output and revenue are given by

$$q^a(\phi) = ((1+\delta)^{\sigma} - 1)q^b(\phi) \quad \text{and} \quad r^a(\phi) = \lambda^r r^b(\phi) \tag{7}$$

[9] While these effects reflect the net effects of upgrading, the model abstracts from embodied emissions generated as a byproduct of production of the upgraded technology.

[10] From a theoretical point of view, expression (6) is a generalization of the Cobb-Douglas pollution-production technology adopted by Copeland and Taylor (2003), in which α can be interpreted as the cost share of non-pollution inputs (Shapiro and Walker 2015). See Andersen (2016) for an extended discussion of the assumption and interpretation.

where $\lambda^r \equiv \left((1+\delta)^{\sigma-1} - 1\right) > 0$ is the percentage increase in revenue associated with the technology upgrade relative to baseline revenue. Expression (7) demonstrates that additional output and revenue are proportional to output and revenue generated by the baseline technology. Similarly, pollution emissions are given by

$$z^b(\phi) = \frac{q^b(\phi)}{\phi^\alpha} \quad \text{and} \quad z^a(\phi) = (\Gamma - 1)z^b(\phi) \tag{8}$$

where $\Gamma = \gamma(1+\delta)^{\sigma-\alpha} > 0$. Expression 8 demonstrates that the technology upgrade reduces pollution emissions, that is $z^a(\phi) < 0$ whenever $\Gamma < 1$.

2.3.1 Fixed and Upgrading Costs

All firms in the market have a fixed production and distribution cost wf, where f is the input requirement in the fixed investment and w is the corresponding wage rate. Firms adopting the technology upgrade have additional input requirement and incur an additional cost associated with financing the investment. Specifically, let $\lambda^c > 0$ represent the percentage increase in fixed costs associated with the additional input requirement (relative to baseline fixed costs), and $b > 1$ represent one plus the interest rate paid on the loan ($b - 1$ is the interest rate). Firms adopting the technology upgrade therefore have an additional input requirement $\lambda^c f$ and incur an additional cost $\lambda^c bwf$. That the additional financing cost bears only on upgrading costs reflects, in a stylized fashion, the insight of the corporate finance literature, which posits that firms use less-costly internal funds first (to finance the baseline technology) and more-costly external funds thereafter (to finance the upgraded technology).[11]

The assumption that upgrading costs are recurring every period retains tractability of the model and adheres to the existing literature that incorporates technology upgrading in the context of firm heterogeneity (Bustos 2011). Because the results are interpreted as long-run effects, this assumption is without loss of generality.[12]

Define net revenue as revenue less variable input costs, and profits as net revenue less fixed input costs. Baseline profits are therefore net revenue less fixed input costs:

$$\pi^b = \frac{r^b}{\sigma} - wf \tag{9}$$

while additional profits are given by

[11] A well-established result in the corporate finance literature posits that, in order to minimize the cost of investment in the context of imperfect information, firms use internal funds first, which are the least-cost source of funds, and use external funds thereafter (Myers and Majluf 1984). While the particular modelling assumption is made for simplicity, the results would be similar under the more general assumption that upgrading increases the share of borrowed funds, implying that an increase in the cost of borrowing would increase the cost of upgrading to a greater extent than it increases the cost of the baseline technology, thus discouraging investment in the technology upgrade.

[12] If, on the other hand, technology upgrading is an upfront, irreversible investment then the effects of changes in default risk would generally be dissimilar among incumbent and entering firms as incumbent firms would be less responsive to changes in the cost of upgrading. Consequently, the short-run effects would generally be smaller in magnitude than the long-run effects of changes in default risk. However, these differences would vanish in the long-run as new entrants supplant incumbent firms.

$$\pi^a = \pi^u - \pi^b = \frac{\lambda^r r^b}{\sigma} - b\lambda^c wf \tag{10}$$

Because net revenue is proportional to revenue, λ^r can also be interpreted as the percentage increase in net revenue relative to baseline net revenue.

Baseline profits and additional profits define the zero-baseline profits and zero-additional profits cutoffs as follows:

$$\frac{r^b(\phi^b)}{\sigma} = wf \quad \text{and} \quad \frac{r^b(\phi^u)}{\sigma} = \frac{b\lambda^c wf}{\lambda^r} \tag{11}$$

where ϕ^b is the cutoff productivity for producing in the market, while ϕ^u is the cutoff productivity for investing in the technology upgrade. Expressions (4), (7), and (11) imply that the ratio of the cutoff productivity for producing in the market to the cutoff productivity for investing in the technology upgrade is given by the following:

$$\frac{\phi^b}{\phi^u} = \left(\frac{\Lambda}{b}\right)^{\frac{1}{\sigma-1}} \tag{12}$$

where $\Lambda = \lambda^r/\lambda^c$.

Assumption 1 Define the ratio of additional net revenue to additional fixed costs as the parameter $\Lambda = \lambda^r/\lambda^c$, and assume that $0 < \Lambda < b$.

Assumption 1 is a necessary and sufficient condition such that the cutoff productivity for producing is exceeded by the cutoff productivity for investing in the technology upgrade $\phi^b < \phi^u$. That is, the additional fixed costs, relative to the additional net revenue, is sufficiently large such that only a subset of more productive firms find it profitable to upgrade, which is consistent with the empirical finding that more productive firms upgrade (Bustos 2011).

2.3.2 Default Risk and Borrowing Costs

Until now, firm profits have been defined over the non-default outcome. However, in every period, firm profits are random variables defined over two possible outcomes: non-default and default. Recall that in the non-default state, firms investing in the technology upgrade borrow $(\lambda^c wf)$ and repay $b(\lambda^c wf)$, where $b - 1$ is the equilibrium interest rate paid on the loan. In the default state, firms exit the market and receive zero profits, and in the case that they borrowed, their creditors are repaid an amount less than the amount in the non-default state. I assume that the default state is an absorbing state with exogenous probability $0 < \eta < 1$.[13] From this point forward, I will refer to the probability of the default state as simply default risk.

In the supplementary web appendix A, I demonstrate that (under a set of conditions) the equilibrium borrowing cost is an increasing function of default risk whenever the lender receives a lesser sum in the default state than the non-default state. To maintain generality,

[13] An absorbing state is one in which it is impossible to leave–that is, firms that exit the market cannot reenter the market in later periods.

I use a reduced-form parameter representing the elasticity of borrowing costs with respect to default risk $\epsilon_{b\eta}$.

Lemma 2 *Borrowing costs are an increasing function of the firm's default risk. That is,*

$$\epsilon_{b\eta} = \frac{\partial b/b}{\partial \eta/\eta} > 0 \tag{13}$$

Proof See appendix A. □

2.3.3 Firm Entry and Exit

There is an unbounded pool of potential entrants deciding on paying a fixed market entry cost wf^e, which allows them to draw a random productivity from the common distribution $G(\phi)$. After realizing their productivity, firms decide to either immediately exit or enter the market and produce using either the baseline or upgraded technology.

I assume that ex-ante productivity draws are Pareto distributed, with the lower bound normalized to one, $G(\phi) = 1 - \phi^{-k}$ and $g(\phi) = k\phi^{-(k+1)}$.[14] To ensure that average per-firm output is finite, I assume that $k > \sigma$. With the ex-ante productivity distribution being Pareto, the ex-post productivity distributions of active firms and firms investing in the technology upgrade are Pareto. That is,

$$\mu^b(\phi) = \frac{k}{\phi}\left(\frac{\phi^b}{\phi}\right)^k \text{ and } \mu^u(\phi) = \frac{k}{\phi}\left(\frac{\phi^u}{\phi}\right)^k \tag{14}$$

Free entry implies that the ex-ante present value of expected profits has to be equal to the cost of entering the productivity draw in equilibrium. Recall that there is an infinite number of time periods, and in every period, firms face an exogenous probability of exit (default risk), which implies that

$$\left(\phi^b\right)^{-k}\frac{\bar{\pi}}{\eta} = wf^e \tag{15}$$

where $\bar{\pi}$ is average profits of active firms, and $\left(\phi^b\right)^{-k}$ is the ex-ante probability that the firm produces. The left hand side is the present value of expected profits from the point of view of a potential entrant that does not yet know its productivity, which is equal to the cost of entry in equilibrium.

Denote the mass of firms of taking productivity draws as M^e, among which a subset will exit after drawing (whenever $\phi < \phi^b$) and the remainder will enter the market and take up production (whenever $\phi \geq \phi^b$). Denote the mass of producing firms as M^b, among which a subset are producing using the upgraded technology. Denote the mass of producing firms using the upgraded technology as M^u, implying that the share of producing firms that invest in the technology upgrade is M^u/M^b, where $M^b > M^u$. In a steady-state equilibrium, the mass of new firms entering the market and taking up production is equal to the mass of firms exiting the market, implying that

[14] Eaton et al. (2011), among others, document that firm size approximately follows this distribution.

$$(\phi^b)^{-k} M^e = \eta M^b \tag{16}$$

where the left hand side is the mass of firms taking productivity draws multiplied by the probability of entering the market and taking up production, while the right hand side is the mass of producing firms multiplied by the probability of exiting.

Lemma 3 *Define the probability of upgrading as the probability that a random firm in the market invests in the technology upgrade, which is denoted as $\theta \equiv M^u/M^b$. Then the probability of upgrading is decreasing in default risk. That is,*

$$\theta \equiv \frac{M^u}{M^b} = \left(\frac{\Lambda}{b}\right)^{\frac{k}{\sigma-1}} \quad \text{and} \quad \frac{\partial\theta/\theta}{\partial\eta/\eta} = -\left(\frac{k}{\sigma-1}\right)\epsilon_{b\eta} < 0 \tag{17}$$

Proof Pareto distributed productivities imply that the conditional probability of upgrading is $\theta = (\phi^b/\phi^u)^k$, which in combination with expression (12) and Assumption 1 yields the desired result. □

Note that the probability of upgrading is decreasing in default risk only if the cutoff productivity for producing is exceeded by the cutoff productivity for investing in the technology upgrade, which requires Assumption 1 to hold. If otherwise, then all firms in the economy invest in the technology upgrade, and consequently, changes in default risk do not generate changes in the probability of upgrading. Because the empirical analysis demonstrates that default risk bears on productivity and emissions intensity, Sect. 2.5 demonstrates that this corroborates that Assumption 1 holds.

2.3.4 Labor Market Clearing

Lemma 3 sheds light on the share of upgrading firms in the market, as well as the ratio of the cutoff productivity for producing in the market to the cutoff productivity for investigating the technology upgrade. In turn, the full employment condition solves for the equilibrium mass of firms in the market. In particular, full employment in production, as well as market entry, is written as

$$L = M^e f^e + M^b\left(f + \int_{\phi^b}^{\infty} \frac{q^b}{\phi}\mu^b(\phi)d\phi\right) + M^u\left(f\lambda^c b + \int_{\phi^u}^{\infty} \frac{q^a}{\phi(1+\delta)}\mu^u(\phi)d\phi\right) \tag{18}$$

where L is the exogenous labor supply. Appendix A demonstrates that the employment condition (18) solves for the equilibrium mass of firms in the market as follows:

$$M^b = \frac{L}{\sigma F}\left(\frac{k}{k-(\sigma-1)}\right) \tag{19}$$

where $k/(k-(\sigma-1)) > 0$ is a fixed parameter and $F \equiv f(1+\theta\lambda^c b) > 0$ is average fixed cost paid by firms in the market. Intuitively, because the distribution of productivities can be inferred from average fixed cost, expression (19) demonstrates that average fixed costs and the supply of labor jointly determine the mass of firms (coupled with various fixed model parameters). Default risk increases borrowing costs, and in turn increases average fixed costs, but default risk decreases the probability of upgrading, and in turn decreases average fixed costs. Proposition 1 demonstrates that the magnitude of the elasticity of the

probability of upgrading with respect to default risk exceeds the magnitude of the elasticity of borrowing costs with respect to default risk, implying that average fixed costs are decreasing in default risk. Because there is an exogenous supply of labor, expression (19) sheds light on the relationship between average fixed costs and the mass of firms in the market. In particular, expression (19) demonstrates that average firm size, characterized by average employment per firm (L/M^b), is increasing in and proportional to average fixed costs. Thus, as a consequence of decreasing average fixed costs, default risk is associated with a greater mass of firms in the market but firms that are smaller on average.

2.4 Analysis

The aim of this subsection is to analyze the effect of default risk on aggregate pollution emissions, decomposing the effect of default risk into analytically distinct effects: the market-size, technology-upgrading, and selection effect. In addition to analyzing the effect on aggregate emissions, the effects of default risk on aggregate output and emissions intensity are analyzed, thereby dividing the effect on aggregate emissions according to aggregate scale and technique effects, respectively.[15] Moreover, the extent to which the three effects are manifested through technique and scale effects is also analyzed. Intuitively, aggregate emissions are the product of the mass of firms in the market and average emissions, the latter of which is the sum of average baseline emissions and expected additional pollution emissions (defined as the product of the probability of upgrading and average additional pollution emissions among upgrading firms). In turn, average baseline emissions is dependent on the baseline productivity threshold (selection effect), whereas expected additional pollution emissions is dependent on the upgrading productivity threshold, which in turn bears on the probability of upgrading and average additional pollution emissions among upgrading firms (technology-upgrading effect). The remainder of this section formalizes this decomposition.

First, average pollution emissions and output are given by the following:

$$\bar{z} = \int_{\phi^b}^{\infty} z^b \mu^b(\phi) d\phi + \theta \int_{\phi^u}^{\infty} z^a \mu^u(\phi) d\phi = \bar{z}^b + \theta \bar{z}^a \tag{20}$$

$$\bar{q} = \int_{\phi^b}^{\infty} q^b \mu^b(\phi) d\phi + \theta \int_{\phi^u}^{\infty} q^a \mu^u(\phi) d\phi = \bar{q}^b + \theta \bar{q}^a \tag{21}$$

where $\bar{z}^b$ is average baseline pollution emissions, and θ and $\bar{z}^a$ are the probability of upgrading and additional pollution emissions among upgrading firms, respectively.

Next, aggregate pollution emissions and output are given by the following:

$$Z = M^b \bar{z} \quad \text{and} \quad Q = M^b \bar{q} \tag{22}$$

where Z is aggregate pollution emissions, Q is aggregate output, and M^b is the mass of firms in the market. Because aggregate emissions can be expressed as the product of aggregate emissions intensity and output ($Z = E^* Q$), it follows that aggregate emissions

[15] In contrast to representative-firm models, the effect on aggregate emissions cannot be directly inferred from firm-level scale and technique effects, as default risk bears on both the number and type of firms in the industry, thereby requiring a more general framework to uncover the underlying mechanisms of the model.

intensity is the ratio of average pollution emissions to average output $E = Z/Q = \bar{z}/\bar{q}$. Expression (22) implies that the elasticity of aggregate pollution emissions with respect to default risk is thus the sum of the elasticity of the mass of firms and the elasticity of average pollution emissions. That is,

$$\frac{\partial Z/Z}{\partial \eta/\eta} = \frac{\partial M^b/M^b}{\partial \eta/\eta} + \frac{\partial \bar{z}/\bar{z}}{\partial \eta/\eta} \tag{23}$$

where the first term represents the elasticity of the mass of firms and the second term represents the elasticity of average pollution emissions. Similarly, the elasticity of aggregate output with respect to default risk is the sum of the elasticity of the mass of firms and the elasticity of average output, which in turn decomposes the scale and technique effects of default risk. That is,

$$\frac{\partial Q/Q}{\partial \eta/\eta} = \frac{\partial M^b/M^b}{\partial \eta/\eta} + \frac{\partial \bar{q}/\bar{q}}{\partial \eta/\eta} \quad \text{and} \quad \frac{\partial E/E}{\partial \eta/\eta} = \frac{\partial \bar{z}/\bar{z}}{\partial \eta/\eta} - \frac{\partial \bar{q}/\bar{q}}{\partial \eta/\eta} \tag{24}$$

where the first term represents the scale effect of default risk and the second term represents the technique effect of default risk on aggregate pollution (the effect on aggregate emissions is the sum of the scale and technique effects). Using the above framework, the underlying mechanisms of the model are described in the Propositions and Corollaries below.

Proposition 1 (Market-size effect) *The mass of firms is increasing in default risk. That is,*

$$\frac{\partial M^b/M^b}{\partial \eta/\eta} = -\frac{\partial F/F}{\partial \eta/\eta} > 0 \tag{25}$$

Proof Using $\frac{\partial F/F}{\partial \theta/\theta} = \frac{\partial F/F}{\partial b/b} > 0$ implies that $\frac{\partial F/F}{\partial \eta/\eta} = -\left(\frac{k-(\sigma-1)}{\sigma-1}\right)\left(\frac{\partial F/F}{\partial b/b}\right)\epsilon_{b\eta} < 0$. □

As mentioned, (19) demonstrated that the mass of firms in the market is inversely related to average fixed costs, while the above demonstrates that average fixed cost is decreasing in default risk. The underlying mechanisms can be explained as follows. Consider a reduction in default risk, which in turn, decreases borrowing costs and encourages a subset of firms to invest in the technology upgrade that otherwise would not have invested in the upgrade. Consequently, equilibrium output expands among impacted firms, but residual demand falls across all firms, rendering the least productive firms unprofitable and thus exit the market. A reduction (increase) in default thus reduces (increases) the mass of producing firms.

Corollary 1.1 *The productivity threshold in which firms take up production is decreasing in default risk. That is,*

$$\frac{\partial \phi^b/\phi^b}{\partial \eta/\eta} < 0 \tag{26}$$

Proof See Appendix A. □

Next, I turn to the effects of default risk on average emissions and output.

Proposition 2 (Selection and technology-upgrading effects) *The effect of default risk on average pollution emissions and output can be decomposed according to selection and technology-upgrading effects. That is,*

$$\frac{\partial \bar{z}/\bar{z}}{\partial \eta/\eta} = \frac{\bar{z}^b}{\bar{z}} \frac{\partial \bar{z}^b/\bar{z}^b}{\partial \eta/\eta} + \frac{\theta \bar{z}^a}{\bar{z}} \left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{z}^a/\bar{z}^a}{\partial \eta/\eta} \right) \tag{27}$$

and

$$\frac{\partial \bar{q}/\bar{q}}{\partial \eta/\eta} = \frac{\bar{q}^b}{\bar{q}} \frac{\partial \bar{q}^b/\bar{q}^b}{\partial \eta/\eta} + \frac{\theta \bar{q}^a}{\bar{q}} \left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{q}^a/\bar{q}^a}{\partial \eta/\eta} \right) \tag{28}$$

where the first terms correspond to the selection effects and the second terms correspond to the technology-upgrading effects. While the effect on average pollution emissions is ambiguous, the effect on average output is negative.

Proof Follows from expressions (20) and (21). See appendix A for proof that the effect on average output is negative. □

The intuition for the effect on average output is as follows. An increase in default risk reduces the baseline productivity threshold, thereby resulting in lower average baseline productivity and in turn average baseline output (selection effect). Moreover, it also reduces the probability of upgrading, which reduces expected additional output (technology-upgrading effect). While an increase in default risk also reduces average baseline emissions by reducing the baseline productivity threshold (selection effect), the reduction in the probability of upgrading might increase or decrease expected additional pollution emissions as additional emissions might be negative. Thus, in contrast to the effect on average output, the effect on average pollution emissions is ambiguous as there are potentially confounding effects.

Proposition 3 (Aggregate pollution decomposition) *The effect of default risk on aggregate pollution emissions can be decomposed into (i) market-size, (ii) selection, and (iii) technology-upgrading effects.*

$$\frac{\partial Z/Z}{\partial \eta/\eta} = \frac{\partial M^b/M^b}{\partial \eta/\eta} + \frac{\bar{z}^b}{\bar{z}} \frac{\partial \bar{z}^b/\bar{z}^b}{\partial \eta/\eta} + \frac{\theta \bar{z}^a}{\bar{z}} \left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{z}^a/\bar{z}^a}{\partial \eta/\eta} \right) \tag{29}$$

where the first term corresponds to the market size effect, the second term corresponds to the selection effect, and the third term corresponds to the technology-upgrading effect.

Proof Follows from expressions (23) and (27). □

Because the effects of default risk on aggregate pollution emissions are confounding in nature (the first is positive, the second is negative, and the third is ambiguous), the aggregate effect on pollution emissions is ambiguous.

Corollary 3.1 (Scale and technique effects decomposition) *The scale effect (the effect of default risk on aggregate output) can be decomposed according to (i) market-size, (ii) selection, and (iii) technology-upgrading effects. That is,*

$$\frac{\partial Q/Q}{\partial \eta/\eta} = \frac{\partial M^b/M^b}{\partial \eta/\eta} + \frac{\bar{q}^b}{\bar{q}}\frac{\partial \bar{q}^b/\bar{q}^b}{\partial \eta/\eta} + \frac{\theta \bar{q}^a}{\bar{q}}\left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{q}^a/\bar{q}^a}{\partial \eta/\eta}\right) \tag{30}$$

where the first term corresponds to the market size effect, the second term corresponds to the selection effect, and the third term corresponds to the technology-upgrading effect. The technique effect (the effect of default risk on aggregate emissions intensity) on aggregate emissions intensity can be decomposed according to (i) selection and (ii) technology-upgrading effects. That is,

$$\begin{aligned}\frac{\partial E/E}{\partial \eta/\eta} = &\left[\frac{\bar{z}^b}{\bar{z}}\frac{\partial \bar{z}^b/\bar{z}^b}{\partial \eta/\eta} - \frac{\bar{q}^b}{\bar{q}}\frac{\partial \bar{q}^b/\bar{q}^b}{\partial \eta/\eta}\right] \\ &+ \left[\frac{\theta \bar{z}^a}{\bar{z}}\left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{z}^a/\bar{z}^a}{\partial \eta/\eta}\right) - \frac{\theta \bar{q}^a}{\bar{q}}\left(\frac{\partial \theta/\theta}{\partial \eta/\eta} + \frac{\partial \bar{q}^a/\bar{q}^a}{\partial \eta/\eta}\right)\right]\end{aligned} \tag{31}$$

where the first term represents the net selection effect and the second term represents the net technology-upgrading effect.

2.5 From Theory to Estimation

The primary objective of this paper is to determine the effect of economy-wide default risk on aggregate pollution emissions, and to decompose the effect along the lines of Propositions 2 and 3, and Corollary 3.1.[16] To this end, this paper estimates the elasticity of productivity and emissions intensity with respect to default risk. As demonstrated below, these elasticities represent composite parameters of the model rather than individual parameters. However, these composite parameters, coupled with parameters from previous studies, are sufficient statistics to numerically quantify the results in the model. Because only firms operating in the market are observed ($\phi \geq \phi^b$), productivity of producing firms is given by:

$$\phi^* = \begin{cases} \phi & \text{if } \phi < \phi^u \\ \phi(1+\delta) & \text{if } \phi \geq \phi^u \end{cases} \tag{32}$$

Because technology choice is not observable in the dataset, productivity is a random variable with conditional expectation equal to

$$E[\phi^*] = \phi(1-\theta) + \phi(1+\delta)\theta = \phi(1+\delta\theta) \tag{33}$$

where θ is the probability that a firm invests in the technology upgrade as defined in Lemma 3.

Similarly, emissions intensity of producing firms is given by (see equation 6):

$$e^* = \begin{cases} \frac{1}{\phi^\alpha} & \text{if } \phi < \phi^u \\ \frac{\gamma}{((1+\delta)\phi)^\alpha} & \text{if } \phi \geq \phi^u \end{cases} \tag{34}$$

[16] To be precise, the objective is to numerically quantify expressions (29), (30), and (31), and to decompose the expressions according to the (i) market-size, (ii) selection, and (iii) technology-upgrading effects.

Consequently, emissions intensity is also a random variable with conditional expectations equal to

$$E[e^*] = \left[\frac{1}{\phi^\alpha}\right](1-\theta) + \left[\frac{\gamma}{((1+\delta)\phi)^\alpha}\right]\theta = \frac{1}{\phi^\alpha}(1+\tilde{\gamma}\theta) \tag{35}$$

where $\tilde{\gamma} = \gamma(1+\delta)^{-\alpha} - 1$.

Expressions (33) and (35) generate the reduced-form specifications used to estimate the elasticity of productivity and emissions intensity with respect to default risk. In particular, the differential (in percentage terms) of productivity and emissions intensity can be expressed in terms of their total differentials, which are functions of the differentials (in percentage terms) of the independent variables. That is,

$$\begin{aligned}\frac{dE[\phi^*]}{E[\phi^*]} = &-\left[\frac{\delta\theta}{1+\delta\theta}\left(\frac{k}{\sigma-1}\right)\epsilon_{b\eta}\right]\frac{d\eta}{\eta} + \frac{d\phi}{\phi} \\ &+\left[\frac{\delta\theta}{1+\delta\theta}\left(\frac{k}{\sigma-1}\right)\right]\frac{d\Lambda}{\Lambda} + \left(\frac{\delta\theta}{1+\delta\theta}\right)\frac{d\delta}{\delta}\end{aligned} \tag{36}$$

which implies that the elasticity of productivity with respect to default risk is given by the following:

$$\frac{dE[\phi^*]/E[\phi^*]}{d\eta/\eta} = -\frac{\delta\theta}{1+\delta\theta}\left(\frac{k}{\sigma-1}\right)\epsilon_{b\eta} < 0 \tag{37}$$

Similarly, the differential (in percentage terms) of emissions intensity is given by:

$$\begin{aligned}\frac{dE[e^*]}{E[e^*]} = &-\left[\frac{\tilde{\gamma}\theta}{1+\tilde{\gamma}\theta}\left(\frac{k}{\sigma-1}\right)\epsilon_{b\eta}\right]\frac{d\eta}{\eta} - \alpha\frac{d\phi}{\phi} \\ &+\left[\frac{\tilde{\gamma}\theta}{1+\tilde{\gamma}\theta}\left(\frac{k}{\sigma-1}\right)\right]\frac{d\Lambda}{\Lambda} + \left(\frac{\tilde{\gamma}\theta}{1+\tilde{\gamma}\theta}\right)\frac{d\tilde{\gamma}}{\tilde{\gamma}}\end{aligned} \tag{38}$$

which implies that the elasticity of emissions intensity with respect to default risk is given by the following:

$$\frac{dE[e^*]/E[e^*]}{d\eta/\eta} = -\frac{\tilde{\gamma}\theta}{1+\tilde{\gamma}\theta}\left(\frac{k}{\sigma-1}\right)\epsilon_{b\eta} \tag{39}$$

By Lemma 1, the technology upgrade reduces emissions intensity whenever $\gamma^{1/\alpha} < 1+\delta$, which implies that $\tilde{\gamma} < 0$. Intuitively, default risk reduces the likelihood of investing in the technology upgrade, implying that (39) is positive if and only if the technology upgrade reduces emissions intensity.

Equations (33) and (35) demonstrate that default risk bears on firm-level productivity and emissions intensity if and only if default risk bears on the probability of upgrading. Thus, the empirical finding that default risk is a significant determinant of productivity and emissions intensity (as documented below) supports the assumption that a subset of firms are not investing in the technology upgrade (at extant levels of borrowing costs), thereby corroborating Assumption 1. Moreover, the empirical finding also corroborates the modelling assumption that default risk bears on the fixed cost of the upgraded technology to a greater extent than it bears on the fixed cost of the baseline technology (otherwise default risk would not affect the relative cost of the upgraded technology and in turn affect upgrading decisions).

3 Empirical Analysis

3.1 Data Sources and Description

This paper uses establishment-level panel data for manufacturing facilities in the United States. To this end, I combine two primary datasets. First, I use the Environmental Protection Agency's (EPA) Risk-Screening Environmental Indicators (RSEI) as a comprehensive measure of establishment-level pollution emissions. Second, I merge the RSEI dataset with establishment-level data from Dunn and Bradstreet's (D&B) National Establishment Time Series (NETS).[17] The NETS dataset contains information on establishment-level annual sales, employment, and other characteristics. Moreover, the NETS also contains establishment-level measures of creditworthiness, compiled by D&Bs DUNS Marketing Information archive.

3.1.1 Pollution Emissions

The RSEI aggregates chemical release data from the EPA's Toxic Release Inventory (TRI) to assess the aggregate damages caused by an establishment's pollution emissions. The TRI has become the primary source by which researchers, regulators, and environmentalists, assess establishment-level environmental performance.[18] Among the three measures of aggregate pollution emissions, this paper focuses on 'Hazard' emissions, which is the weighted sum of all chemical releases, where the weights correspond to the toxicity level (as measured by epidemiological studies) of the chemical released.

3.1.2 Establishment Characteristics

The NETS dataset contains annual establishment-level measures of creditworthiness from D&B, called PayDex Scores, which range between 0 and 100 in ascending order of creditworthiness.[19] The primary advantage of PayDex Scores is that they are the most extensively-used measures of business credit, covering both private and publicly-traded firms. The NETS dataset also contains information on the age of the establishment (year start) and the year in which the establishment went out of business (if applicable). Equipped with this information, it is possible to verify if in fact PayDex scores are related to default risk, and to construct measures of default risk, which I discuss in the subsequent section.

The NETS dataset contains information on establishment-level sales, employment, industry classification, and various other characteristics. Using these variables, I construct measures of output by deflating establishment-level sales using industry price deflators

[17] The NETS database is proprietary data compiled by Walls and Associates and covers all establishments in the United States starting in 1990.

[18] While it is beyond the scope of this paper to address the various concerns of the TRI, de Marchi (2006) investigate the accuracy of the TRI and find evidence of misreporting in two, of the twelve investigated, chemicals.

[19] According to D&B, PayDex Scores is a measure of both payment history and default risk, where a score above 80 indicates low risk and below 50 indicates high risk. The model used by D&B to generate scores is proprietary, and the company does not provide information regarding the interpretation of PayDex Scores in terms of the corresponding default probability. For more information on PayDex Scores, see http://payde x.net/.

from Bartelsman and Gray (1996), and productivity by dividing output by employees.[20] One drawback to the NETS database is that it does not include information on hours worked as well as capital stocks, precluding estimating total factor productivity.[21]Neumark et al. (2011) investigate the quality of the NETS data and find that the accuracy of the employment data is of similar quality as the Current Population Survey (CPS) and the Current Employment Statistics (CES) Payroll data.

3.2 Sample of Analysis and Summary Statistics

The data consists of an unbalanced panel of manufacturing plants in 2-digit Standard Industrial Classification (SIC) 20–39, starting in 1990 and ending in 2009.[22] The RSEI dataset does not distinguish between missing and zero-valued emissions; hence, I exclude all plants with non-positive value of emissions. Finally, because there are very few establishments in the "Tobacco" and "Apparel" sectors (11 and 32, respectively), and no instances of establishments exiting the market in these sectors, I omit all establishments in these sectors because generating a measure of default risk is not possible.[23] The final sample consists of 25,847 establishments and 216,506 establishment-year observations.

Table 1 reports summary statistics by industry (in ascending order of 2-digit SIC). The first column reports the number of unique establishments by sector, while the second column reports the corresponding average number of years establishments are observed in the sample. For example, the "Chemicals" sector consists of 4143 unique establishments, and establishments are observed for 9.36 years on average.

Because pollution emissions and labor productivity are not denoted in meaningful units, both measures are normalized to fit a standard normal distribution for comparison. Specifically, Emissions is the ratio of hazard pollution emissions to deflated Sales, while Productivity is the ratio of deflated sales to the number of employees. The Table demonstrates that both variables exhibit significant variation, both within and between sectors. Sales is establishment-level sales (divided by 1 million) in constant 2014 US dollars (USD). Average sales is approximately 55 million USD and the standard deviation is around 155 million USD, demonstrating that the distribution of sales are positively skewed. The mean Credit Score is 74.6 and the corresponding standard deviation is 6.4. Recall, D&B classify plants with Credit Scores above 80 as being low risk of default, suggesting that most establishments have at least some risk of default. Default risk is defined and described in the subsequent section.

[20] Establishment-level prices are unavailable and would be endogenous, thereby complicating the analysis.

[21] Labor productivity might vary across industries due to variation in capital intensity. However, the baseline analysis uses log labor productivity and accounts for establishment fixed effects, which implies that variation in productivity is in terms of percentage changes from an establishment's mean labor productivity. Moreover, the baseline analysis accounts for industry by year fixed effects, and the coefficient estimates do not vary systematically across industries (results available upon request).

[22] The start and end years of the panel are determined by the availability of the NETS database.

[23] Few establishments in these sectors report pollution emissions because they tend to be the least polluting among the manufacturing sector. It is therefore not particularly consequential to omit these establishments.

Table 1 Summary statistics: mean and (standard deviation)

	#Est.	#Years	Emissions	Productivity	Sales (mil)	Credit score	Default risk (%)
Food	1624	7.60	− 0.75 (0.71)	− 0.04 (1.07)	65.49 (112.69)	77.48 (4.91)	0.20 (0.26)
Textiles	230	8.56	− 0.47 (0.80)	− 0.11 (0.71)	46.21 (82.48)	75.52 (6.45)	0.34 (0.49)
Lumber	887	7.27	− 0.01 (0.91)	− 0.26 (0.80)	27.25 (36.82)	78.02 (5.50)	0.25 (0.33)
Furniture	616	6.75	− 0.52 (0.76)	− 0.54 (0.60)	46.53 (108.91)	75.85 (6.56)	0.21 (0.30)
Paper	737	8.12	− 0.22 (0.81)	− 0.10 (0.84)	60.55 (96.49)	75.52 (5.85)	0.15 (0.20)
Printing	267	6.22	− 0.60 (0.60)	− 0.34 (0.69)	43.58 (67.54)	76.44 (6.36)	0.37 (0.40)
Chemicals	4,143	9.36	− 0.10 (0.86)	0.15 (0.89)	36.03 (94.98)	74.18 (6.18)	0.19 (0.24)
Petroleum and coal	474	8.70	0.13 (0.91)	− 0.19 (1.25)	40.73 (171.43)	75.28 (6.22)	0.25 (0.30)
Rubber and plastics	1,774	8.84	− 0.41 (0.77)	− 0.27 (0.66)	27.59 (41.72)	74.53 (6.59)	0.21 (0.25)
Leather	96	7.53	0.44 (1.44)	− 0.27 (0.76)	29.75 (33.37)	75.90 (6.59)	0.84 (0.77)
Stone, clay, and glass	1,224	7.81	− 0.19 (1.00)	− 0.07 (0.77)	30.31 (46.63)	75.84 (6.00)	0.16 (0.24)
Primary metals	2,491	9.63	0.50 (0.96)	0.06 (0.88)	45.74 (107.09)	74.81 (6.38)	0.27 (0.35)
Fabricated metals	3,859	8.63	0.39 (1.02)	− 0.50 (0.95)	22.22 (40.77)	74.47 (6.72)	0.18 (0.25)
Industrial machinery	2,119	8.14	0.41 (1.15)	− 0.05 (0.94)	62.02 (136.70)	73.50 (6.13)	0.24 (0.30)
Electronics	2,465	7.54	− 0.12 (0.85)	0.73 (1.45)	87.63 (190.60)	73.11 (6.31)	0.31 (0.39)
Transportation equip.	1,691	8.13	0.08 (1.07)	0.43 (0.72)	180.39 (392.12)	72.88 (6.93)	0.26 (0.31)
Instruments	750	7.00	− 0.23 (1.03)	0.49 (0.84)	126.81 (289.23)	73.84 (5.64)	0.27 (0.38)
Misc. manufacturing	400	7.23	− 0.35 (0.93)	0.13 (0.98)	52.19 (90.68)	75.35 (6.62)	0.39 (0.46)
Total	25,847	8.38	− 0.00 (1.00)	0.00 (1.00)	55.06 (156.99)	74.63 (6.40)	0.23 (0.31)

3.3 Default Risk

Estimating the elasticity of productivity and emissions intensity with respect to default risk requires a measure of default risk in terms of the probability of default, which is not available at the establishment or firm level. The aim of this section is to generate an appropriate measure of default risk using information on the year in which establishments go out of business (exit the market) and variation in credit scores across establishments. To this end, I estimate the following pooled binary choice model:

$$\text{Prob}(\text{Death}_{ist+1} = 1|\mathbf{x}_{ist}) = \mathbf{\Phi}(\alpha_t + \alpha_s + \alpha_1 \text{Credit Score}_{ist}) \tag{40}$$

where i indexes establishments, s indexes sectors, and t indexes years. The dependent variable, Death_{ist+1}, is a binary variable equal to 1 if the firm exits the market in the subsequent period $t + 1$.[24] The independent variables include year and 3-digit SIC-industry fixed effects and Credit Score. The cumulative distribution function $\mathbf{\Phi}$ is assumed to be normal or logistic, giving rise to the probit and logit model, respectively.

The 3-digit SIC-industry fixed effects account for default-risk heterogeneity across narrowly defined industries (345 industries), while the year fixed effects account for common time-varying factors, such as technology and demand shocks.[25] Credit Score is included to account for variation in default risk within industries and over time.

In many instances, the choice between probit and logit does not make much difference. However, because the difference between the two distributions is primarily in the tails (logistic distribution has heavier tails), the two models might generate different predictions when the dependent variable is highly uneven. Because there are only 655 instances of firm death among 25,847 (2.5%) in the sample of analysis, the two models might give substantively different results and, on theoretical grounds, it is difficult to justify one model or another. Therefore, both models are used.

Figure 1 presents the distribution (univariate kernel density estimation) of credit scores among non-surviving ($\text{Death}_{ist+1} = 1$) and surviving ($\text{Death}_{ist+1} = 0$) establishments. While both distributions are skewed to the left, the distribution of non-surviving establishments exhibits more negative skewness as the left tail is relatively fatter. That is, non-surviving establishments are more likely to have low credit scores relative to surviving firms.

Table 2, columns (1) and (3), report the baseline probit and logit regressions results, respectively. As anticipated, Credit Score is inversely related to Death, and the relationship is statistically significant at all conventional levels of significance. Table 2 also reports average marginal effects (elasticities), which indicate that a 1% increase in Credit Score reduces the probability of Death by 2.8% and 2.6% in the probit and logit regressions, respectively.[26] Columns (2) and (4) are discussed in the subsequent section.

[24] Firms exiting in period t do not report in that period, precluding estimation of a contemporaneous model. Exiting the market is an "absorbing" state as there are no instances of establishments taking up production in subsequent years.

[25] Alternative specifications also control for 4-digit SIC industry fixed effects and industry by year fixed effects. However, maximum likelihood estimation drops almost all of the additional fixed effects because Death is a relatively rare event, and consequently there is no variation in the dependent variable within these fixed effects categories.

[26] Similarly, the odds ratio in column (3) is 0.989, implying that a 1 unit increase in Credit Score reduces the probability of Death by around 1.1%.

The primary purpose of this section is to assign default risk to establishments using the predicted values generated from the regression given in (40). To this end, I define

$$\text{Default Risk}_{ist} = \text{E}[\text{Death}_{ist+1}|\mathbf{x}_{ist}] \tag{41}$$

The interpretation of Default Risk$_{ist}$ is the predicted probability that the establishment will exit the market (default) in a given year, which is consistent with the theoretical model. The generated variable is, however, measured with error, which has implications that are discussed in the subsequent section.

Figure 2 plots the distribution (univariate kernel density estimation) of (log) default risk generated from the probit and logit regressions. The figure demonstrates that the two models generate nearly identical default risk distributions, suggesting that the choice between probit and logit should not make much difference.

The last two columns in the summary statistics table (Table 1) report the mean and standard deviation of default risk across sectors.[27] The mean default risk among all establishments in the sample is around 0.2%, and the standard deviation is around 0.3%.[28] Default risk varies across sectors, ranging from 0.15% to over 0.80%, as well as within sectors, as the standard deviation within sectors is similar to the overall standard deviation.

Figure 3 plots the mean default risk by year (solid line) and the corresponding 90% confidence interval over the 1990–2009 period. The annual exit rate (percentage of establishments exiting the market by year) is also plotted (dash line with point marker). The figure demonstrates that the actual exit rate is highly consistent with the number of establishment deaths predicted by the regression model. The figure also demonstrates that default risk varies systematically, and quite significantly in percentage terms, over time, ranging from less than 0.05% in the mid-1990s and 2000s to over 0.5% in the early 2000s. While default risk predicts firm exit in the run-up to the Great Recession, that the exit rate did not increase more substantially in 2008 and 2009 demonstrates that polluting firms were relatively less affected by the recession than other industries (e.g., financial services and real estate).[29]

3.4 Productivity and Emissions Intensity

The theoretical model generates the empirical specifications used to estimate the elasticity of productivity and emissions intensity with respect to default risk. Recall that productivity is determined by η, ϕ, Λ, and δ, while emissions intensity is determined by η, ϕ, Λ, and $\tilde{\gamma}$. The empirical strategy is to account for the technological parameters, Λ, δ, and $\tilde{\gamma}$, using flexible industry by year fixed effects. Moreover, establishment fixed effects are used to account for the productivity draw ϕ. Because the coefficients in (36) and (38) represent elasticities, the regression specifications employ log-transformed variables. That is,

[27] For the remainder of this section, default risk is generated using the probit model. The results are similar using the logit model, and are available upon request.

[28] While the mean default risk is less than the manufacturing sector in general, this observation is not unexpected as the sample is not necessarily representative of the manufacturing sector and polluting establishments tend to be larger and more capital intensive on average.

[29] One explanation is that polluting firms tend to be large and more capital intensive, implying they would be less likely to exit than firms that are less capital intensive and thus have relatively less fixed costs. Moreover, because the sample of analysis ends in 2009, many of the effects of the recession might not have yet been realized.

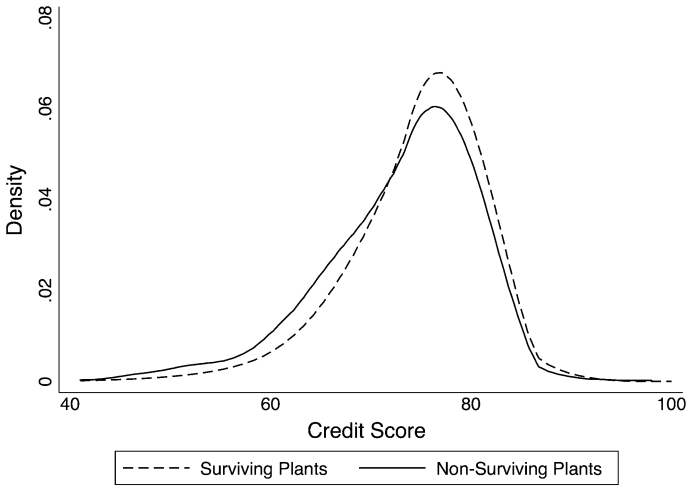

Fig. 1 Kernel density of credit scores. *Note* The figure plots the kernel density distributions of Credit Score for surviving ($\text{Death}_{ist+1} = 0$) and non-surviving ($\text{Death}_{ist+1} = 1$) establishments

Table 2 Probit/logit analysis of firm death

	Probit		Logit	
	(1)	(2)	(3)	(4)
Credit score	− 0.0112†	− 0.0419†	− 0.0352†	− 0.1241†
	(0.0027)	(0.0081)	(0.0080)	(0.0236)
Credit score × public facility		0.0236†		0.0686†
		(0.0059)		(0.0173)
Public facility		− 1.6613†		− 4.8172†
		(0.4377)		(1.2838)
Marginal effects (elasticity)				
Credit score	− 2.7875†	− 10.4107†	− 2.6274†	− 9.2502†
Credit score × public facility		7.6764†		6.7209†
Public facility		− 7.2348†		− 6.3314†
Log likelihood	− 3198.03	− 3186.97	− 3196.55	− 3185.95
Observations	216,506	216,506	216,506	216,506
Industry dummies	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes

The dependent variable is equal to 1 if the firm exits the market in the next period ($\text{Death}_{ist+1} = 1$), and 0 if otherwise. Standard errors are in parenthesis

Significance levels: *0.10, **0.05, and †0.01

$$\ln \text{Productivity}_{ist} = \beta_1 \ln \text{Default Risk}_{ist} + \beta_i + \beta_{st} + \epsilon_{ist} \tag{42}$$

and

$$\ln \text{Emissions Intensity}_{ist} = \zeta_1 \ln \text{Default Risk}_{ist} + \zeta_i + \zeta_{st} + \varepsilon_{ist} \tag{43}$$

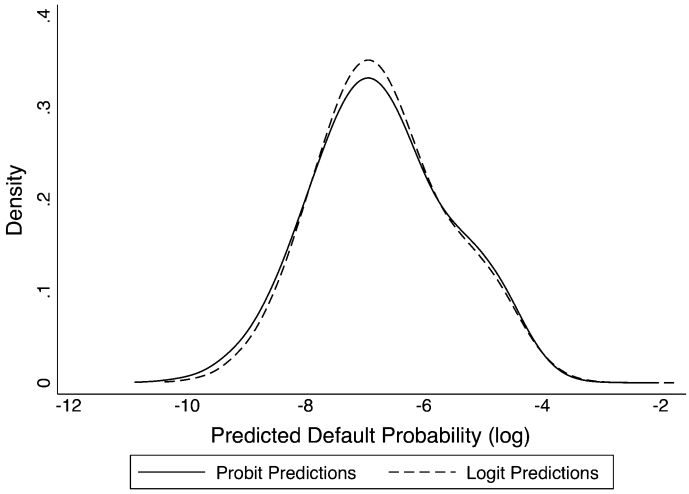

Fig. 2 Kernel density of predicted default probabilities for probit and logit estimations. *Note* Predicted probabilities are generated from the model estimated in Table 2, columns 1 and 3

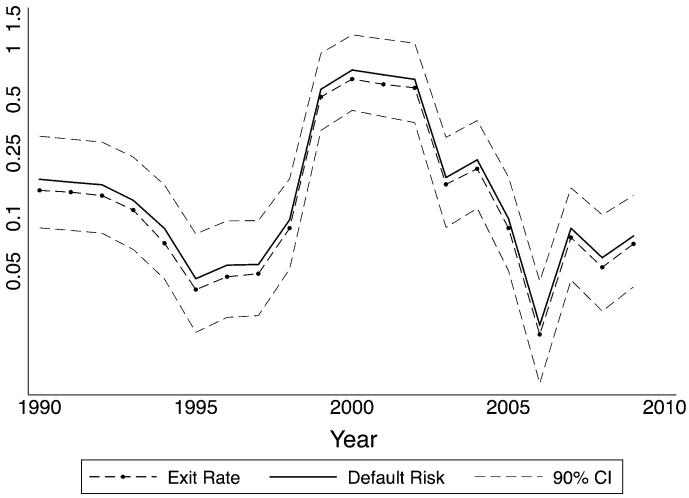

Fig. 3 Exit rate and default risk time series. *Note* The Exit Rate is the number of establishments exiting the market as a percentage of the total number of establishments in the market for a given year. Default Risk is average predicted Default Risk generated from the regression reported in Table 2 column 1

where (as before) i indexes establishments, s indexes sectors, and t indexes years. Establishment fixed effects are captured by the parameters β_i and ζ_i, while industry by year fixed effects are captured by the variables β_{st} and ζ_{st}. The establishment fixed effects account for establishment unobserved heterogeneity, while the industry by year effects account for

aggregate industry time effects.[30] Because the panel is short, estimation is by Fixed Effects where the establishment-specific effects β_i and ζ_i are eliminated using a within transformation.[31] The parameter β_1 is the elasticity of productivity with respect to default risk and ζ_1 is the elasticity of emissions intensity with respect to default risk.[32]

The proposed two-step procedure using unobserved regressors generated from a first-stage auxiliary econometric model has been analyzed by Murphy and Topel (1985) and Pagan (1984). Consistent estimates of the second-stage parameters (β_1 and ζ_1) requires (i) consistent estimation of the first-stage parameters and (ii) strict exogeneity of the regressors in the second-stage.[33] Because Default Risk is imputed from the first-stage model, strict exogeneity implies that Credit Score is uncorrelated with the random components ϵ_{ist} and ε_{ist}. That is, Credit Score is not directly related to Productivity and Emissions Intensity, it is only related to the dependent variables through Default Risk, which I discuss in more detail below.

3.4.1 Regression Results

Columns (1) and (4) of Table 3 report the baseline regression results using Productivity (Panel A) and Emissions Intensity (Panel B) as dependent variables. Columns (1) and (4) correspond to Default Risk generated in Table 2 columns (1) and (3), respectively. The standard errors are cluster-robust standard errors that cluster on industry (bootstrap estimates of the standard errors are reported in the subsequent section). The results indicate that the elasticity of labor productivity with respect to default risk is negative (as expected), while the elasticity of emissions intensity with respect to default risk is positive (consistent with upgrading reducing emissions intensity). In particular, the elasticity of labor productivity is − 0.16 and − 0.17, while the elasticity of emissions intensity is 0.89 and 0.93 in specifications (1) and (4), respectively. Thus, the coefficient estimates are similar across first-stage model specifications (probit and logit), and are significant at conventional significance levels.

Recall one threat to consistency is that (within-establishment) variation in Credit Score might be linked to variation in Productivity and Emissions Intensity through channels unrelated to default risk. While variation in Credit Score is arguably not directly linked to variation in Productivity and Emissions Intensity, both might be linked to time-varying omitted factors, such as variation in managerial skill. For example, hiring a high-skilled manager might increase productivity and reduce pollution emissions, as well as increase Credit Score, which would lead to inconsistent estimates.

[30] The results are robust to the inclusion of state by year fixed effects, which are available upon request.

[31] The within transformation implies that the dependent and independent variables are differences from the mean. The establishment-specific effects β_i and ζ_i can also be eliminated using a first-difference specification, wherein the dependent and independent variables are differences between current and lagged periods. The results are consistent using this alternative approach and the coefficients and standard errors are provided in the notes of Table 3.

[32] Recall from equations (36) and (38), $\beta_1 = -(\delta\theta/(1+\delta\theta))(k/(\sigma-1))\epsilon_{b\eta}$ and $\zeta_1 = -(\tilde{\gamma}\theta/(1+\tilde{\gamma}\theta))(k/(\sigma-1))\epsilon_{b\eta}$.

[33] While the two-step procedure generates consistent estimates of the parameters in the second-stage regression, it is well known that failing to account for the presence of imputed regressors yields inconsistent estimates of their standard errors. To remedy this issue, this study employs a two-step bootstrapping procedure to correct the standard errors in the estimated elasticities as well as the numerical model results, which I discuss in more detail in the subsequent section.

Table 3 Productivity and emissions intensity elasticities

	Probit first-stage			Logit first-stage		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Productivity						
Default risk	– 0.1582†	– 0.1013†	– 0.1542†	– 0.1733†	– 0.1156†	– 0.1941†
	(0.0181)	(0.0098)	(0.0162)	(0.0175)	(0.0105)	(0.0190)
Credit score			– 0.0044†			– 0.0055†
			(0.0007)			(0.0008)
Adj. R-sq	0.128	0.125	0.128	0.128	0.126	0.129
Observations	216,506	216,506	216,506	216,506	216,506	216,506
Panel B: Emissions intensity						
Default risk	0.8866†	0.5463†	0.9563†	0.9304†	0.6066†	1.1808†
	(0.0840)	(0.0570)	(0.0711)	(0.0895)	(0.0665)	(0.0859)
Credit score			0.0344†			0.0401†
			(0.0040)			(0.0043)
Adj. R-sq	0.024	0.023	0.025	0.024	0.023	0.025
Observations	216,506	216,506	216,506	216,506	216,506	216,506
Industry × year dummies	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variables are log Labor Productivity (deflated sales divided by employees) in Panel A, and log (Hazard) pollution emissions intensity (pollution emissions divided by deflated sales) in Panel B. Specifications (1)–(3) use Default Risk predicted using a first-stage probit model (Table 2, columns 1 and 2), while specifications (4)–(6) use Default Risk predicted using a first-stage logit model (Table 2, columns 3 and 4). Specifications (1) and (4) correspond to first-stage regressions reported in columns (1) and (3) of Table 2, while specifications (2) and (3) and (5) and (6) correspond to first-stage regressions reported in columns (2) and (4), respectively. Estimating (1) using a First Difference (instead of Fixed Effects) specification results in point estimates of – 0.1201 (0.0269) and 0.8085 (0.1140) in Panels A and B. The results using First-Differences for specifications (2)–(6) are consistent with the Fixed Effects and available upon request. The estimations use cluster-robust standard errors that are clustered on 3-digit SIC industries, and are in parenthesis

Significance levels: *0.10, **0.05, and †0.01

One approach to address this issue is to exploit variation in the differential impact of Credit Score on Default Risk, and control for Credit Score in the second-stage model. To this end, this paper exploits variation in the impact of Credit Score on Default Risk across publicly-owned and privately-owned establishments (the sample of analysis consists of 32% publicly-owned and 68% privately-owned establishments). In particular, publicly-owned establishments can finance operating costs and investments using retained earnings and equity, while privately-owned establishments are relatively more dependent on external lenders, and credit scores are the yardstick by which lenders assess creditworthiness. Consequently, publicly-owned establishments are less likely to go out of business as a consequence of negative credit shock.

Table 2, columns (2) and (4), report the first-stage probit and logit regression results, respectively, using Credit Score interacted with a dummy variable equal to 1 for

publicly-owned facilities (Public Facility).[34] The results suggest that there is significant variation in the effect of Credit Score across publicly-owned and privately-owned facilities.[35] In particular, the coefficient estimate for Credit Score for privately-owned facilities (− 0.04 and − 0.12) is approximately twice the size of the corresponding estimate for publicly-owned facilities (− 0.02 and − 0.06).[36] Moreover, as expected, publicly-owned facilities are far less likely to go out of business than privately-owned facilities.

Exploiting variation in the differential effect of Credit Score permits controlling for it in the second stage. Consequently, consistency requires, in this case, that the interaction between Credit Score and Public Facility is uncorrelated with the random components in the second stage (ownership status is time invariant in the sample). The threat to consistency is that the differential effects of a credit score shock on variation in default risk might be related to differential effects on variation in Productivity and Emissions Intensity (through channels unrelated to differential effects on default risk). Let's reconsider the example of managerial skill as an omitted variable to clarify this assumption. Suppose, for example, hiring a high-skilled manager improves a facility's credit score to an equal extent for both publicly-owned and privately-owned facilities, but it reduces default risk to a greater extent for privately-owned facilities. Consistency requires that the effect of hiring a high-skilled manager on productivity and emissions intensity should not vary according to ownership status. For example, if hiring a high-skilled manager increases productivity and reduces pollution emissions to a greater extent in privately-owned facilities, and it increases credit scores to the same extent, then the estimates would not be consistent.

Columns (2) and (3) and (5) and (6) of Table 3 report the regression results using the Default Risk generated in columns (2) and (4) of Table 2, respectively. Moreover, columns (2) and (5) do not control for Credit Score, whereas columns (3) and (6) control for Credit Score, in the second-stage regressions. Employing the Public Facility interaction term in the first-stage regressions without controlling for Credit Score in the second-stage generates slightly smaller estimated effects of default risk on labor productivity (Panel A, columns 2 and 5) and emissions intensity (Panel B, columns 2 and 5), while controlling for Credit Score in the second-stage generates slightly larger estimated effects of default on labor productivity (Panel A, columns 3 and 6) and emissions intensity (Panel B, columns 3 and 6).

3.4.2 Alternative Difference-in-Differences Approach

Panel regressions are often subject to criticism for omitting relevant time-varying factors and other endogeneity issues. This section uses an alternative difference-in-differences (DID) approach using variation in Default Risk at the industry level to analyze the elasticity of productivity and emissions intensity with respect to default risk.

Figure 3 demonstrates that Default Risk (and the corresponding exit rate) is subject to large systematic shocks over time. While Default Risk is correlated within years, Default Risk shocks are not uniform across industries as large shocks tend to be particularly acute

[34] Similar to (40), I estimate the following model $\text{Prob}(\text{Death}_{ist+1} = 1|\mathbf{x}_{ist}) = \mathbf{\Phi}(\alpha_t + \alpha_s + \alpha_1 \text{Credit Score}_{ist} + \alpha_2 \text{Credit Score}_{ist} \times \text{Public Facility}_{is} + \alpha_3 \text{Public Facility}_{is})$.

[35] Figures A1 and A2 in the supplementary web appendix B plot the kernel density distributions of default risk for public and private establishments generated using probit (A1) and logit (A2) regressions.

[36] The statistical significance of the interaction term $\text{Credit Score}_{ist} \times \text{Public Facility}_{is}$ demonstrates that the difference between the coefficient estimates is significant at the 1% significance level.

among a subset of industries. For example, average Default Risk increased sharply from 1998 (0.001%) to 1999 (0.060%), representing over a fivefold increase in Default Risk (or a 1.8 increase in log Default Risk), which was the largest Default Risk shock in the period. However, the average change in log Default Risk by industry varied from around 1.46 to 2.21.[37] To exploit variation in the differential impact of the 98/99 Default Risk shock on productivity and emissions intensity, the following model is estimated using ordinary least squares:

$$\Delta \ln \text{Productivity}_{ist} = \tilde{\beta}_1 \Delta \ln \text{Default Risk}_{st}^{-i} + \tilde{\beta} + \tilde{\varepsilon}_{ist} \tag{44}$$

and

$$\Delta \ln \text{Emissions Intensity}_{ist} = \tilde{\zeta}_1 \Delta \ln \text{Default Risk}_{st}^{-i} + \tilde{\zeta} + \tilde{\varepsilon}_{ist} \tag{45}$$

where (as before) i indexes establishments, s indexes sectors, and t indexes years. The dependent variables are (establishment-level) differences in productivity and emissions intensity from 1999 to 1998, and the primary independent variable is the difference in average industry default risk, excluding the default risk of establishment i.[38] The intercept terms $\tilde{\beta}$ and $\tilde{\zeta}$ control for changes in productivity and emissions intensity not due to differential variation in the impact of the 98/99 Default Risk shock. The estimations use cluster-robust standard errors that are clustered on industries.

The primary advantage of the alternative DID approach is its simplicity and potential to overcome the endogeneity issues that arise in panel regressions. The parameters $\tilde{\beta}_1$ and $\tilde{\zeta}_1$ represent the reduced-form elasticity of productivity and emissions intensity with respect to industry-level default risk. Consistency requires that variation in the 98/99 Default Risk shock across industries is uncorrelated with the random components $\tilde{\varepsilon}_{ist}$ and $\tilde{\varepsilon}_{ist}$. That is, variation in the Default Risk shock across industries should be uncorrelated to factors related to changes in productivity and emissions intensity. For example, if industries experiencing more acute Default Risk shocks are subject to less stringent environmental regulations then the assumption would be violated.

The reduced-form results are consistent with the panel regressions, with parameter estimates (standard errors) equal to $\tilde{\beta}_1 = -0.083\ (0.018)$ and $\tilde{\zeta}_1 = 0.511\ (0.210)$.[39] The elasticity of productivity and emissions intensity with respect to default risk can be calculated by dividing the reduced-form parameter estimates by the appropriate-first stage parameters (Wald estimator).[40] Because the elasticity of establishment-level Default Risk with respect to industry-level Default Risk (standard error) is 0.808 (0.002), the elasticity of productivity and emissions intensity with respect to default risk is $-0.083/0.808 = -0.103\ (0.022)$ and $0.511/0.808 = 0.632\ (0.241)$. The Wald estimators generated from the alternative DID approach are highly consistent with the primary results.

[37] For comparison, the average log change in Default Risk over the period is -0.05, while the average absolute-value log change over the period is 0.56.

[38] Precisely, $\Delta \ln \text{Productivity}_{ist} = \ln \text{Productivity}_{is1999} - \ln \text{Productivity}_{is1998}$, and $\Delta \ln \text{Default Risk}_{st}^{-i} = \ln \text{Default Risk}_{s1999}^{-i} - \ln \text{Default Risk}_{s1998}^{-i}$, where $\text{Default Risk}_{s1999}^{-i}$ is the average Default Risk of industry i in year 1999 (and similar for 1998), excluding establishment i.

[39] The corresponding p values are $p = 0.00$ and $p = 0.02$.

[40] In this case, the following first-stage model $\Delta \ln \text{Default Risk}_{st}^{-i} = \tilde{\alpha} \Delta \ln \text{Default Risk}_{ist} + u_{ist}$, is estimated using OLS.

The 98/99 Default Risk shock was selected as a quasi-natural experiment to perform an alternative DID estimation because it generated the largest shock in Default Risk, as well as the most variation across sectors. However, there were secondary Default Risk shocks in 06/07 and 02/03. In particular, average log Default Risk decreased from 2002 to 2003 by 1.34, with industry averages ranging from − 1.65 to − 0.93, while average log Default Risk increased from 2006 to 2007 by 1.17, with industry averages ranging from 0.77 to 1.61. In addition to corroborating the previous results, exploring the second and third largest Default Risk shocks also explores whether the effects of negative and positive Default Risk shocks generate symmetric effects.

Using the same approach to generate the DID estimate for the 98/99 Default Risk shock in all years spanning the sample of analysis, Fig. 4 reports the DID estimates for all years and the corresponding 90% confidence intervals using Productivity and Emissions Intensity as dependent variables in the top and bottom panels, respectively. Figure 4 also includes time series on average industry changes in Default Risk by year (solid line), plotted below the DID estimates in both panels. The series demonstrate that three largest Default Risk shocks occurred in 98/99, 06/07, and 02/03 (in descending order of magnitude), which are indicated using gray vertical bars. In addition, Figure A3 in the supplementary web Appendix B reports the distribution (univariate kernel density estimation) of changes in Default Risk by year, indicating that the three largest Default Risk shocks also generated the largest dispersion in Default Risk across industries.

Figure 4 demonstrates that the Productivity DID estimate (top panel) is consistent in 06/07 Default Risk shock, but is not statistically significant in the 02/03 Default Risk shock.[41] The Emissions Intensity DID estimate (bottom panel) is consistent in the 06/07 and 02/03 Default Risk shocks, but the latter is not statistically significant as the confidence interval contains zero within it's lower bound.[42] Exploiting the three largest Default Risk shocks therefore results in consistent results in the two positive Default Risk shocks, and insignificant results in the one negative Default Risk shock. More generally, the Productivity DID estimate is negative in most years, while the Emissions Intensity DID estimate is positive in most years, with relatively narrower confidence intervals around larger Default Risk shocks.

3.5 Numerical Model

Recall that the theoretical model analyzed the effect of default risk on aggregate pollution emissions, and decomposed the effect according to the underlying mechanisms. The aim of this subsection is to numerically quantify these effects using the elasticities estimated in the previous section.[43] Because many of the structural parameters of the model have been established in the literature, this paper relies on parameter estimates from previous studies. The sources and details of the assumed parameters are provided in Appendix D of the supplementary web appendix.

[41] In particular, the Productivity DID estimate (standard error) is − 0.283 (0.037) in 06/07 and 0.017 (0.016) in 02/03.

[42] In particular, the Emissions Intensity DID estimate (standard error) is 0.653 (0.316) in 06/07 and 0.360 (0.266) in 02/03.

[43] To illustrate the approach using a stylized example, one might estimate the elasticity of labor supply with respect to marginal tax rates, and couple this with an aggregate production function with assumed parameter values, in order to numerically simulate the effect of a change in tax policy on aggregate output.

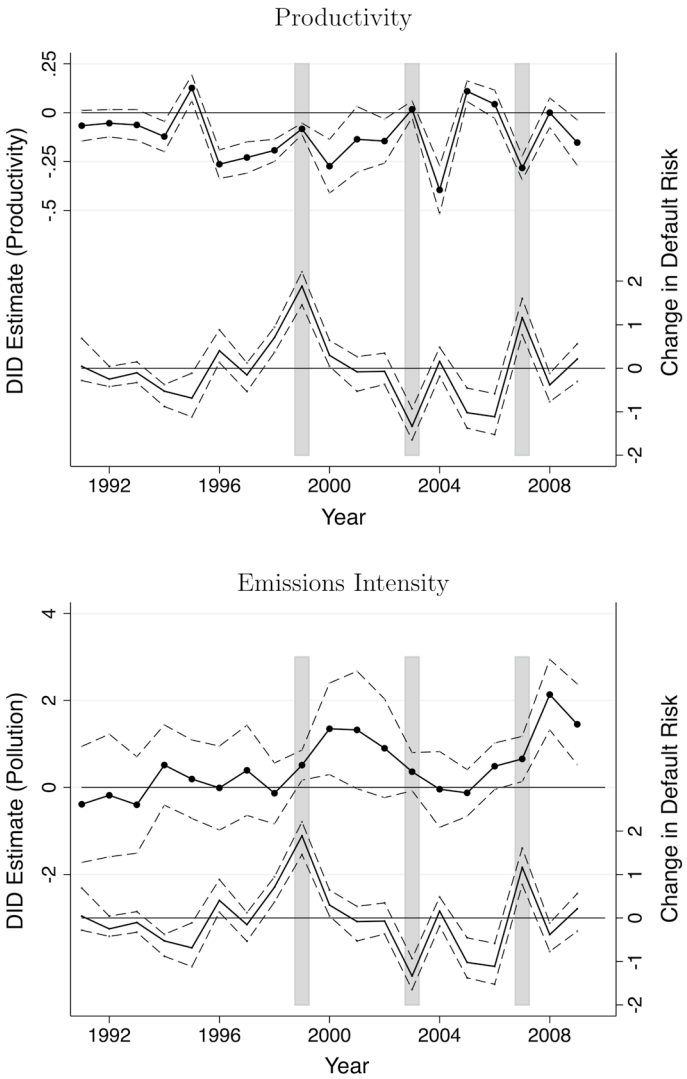

Fig. 4 DID Estimate by Year. *Note* The top (solid) connected line is the DID estimate for (log) Productivity (top panel) and (log) Emissions Intensity (bottom panel) by year and the corresponding 90% confidence interval (dashed lines), where the scale is indicated on the left axis. The bottom (solid) connected line is the log change in average industry Default Risk and the corresponding minimum and maximum average industry Default Risk (dashed lines), where the scale is indicated on the right axis

Table 4 Numerical model results

	Productivity elasticity	Emissions elasticity	Market-size effect
	– 0.1582**	0.8866**	0.1537**
	(0.0705)	(0.4355)	(0.0685)
	Selection effect	Upgrading effect	Aggregate effect
Scale	– 0.0653†	– 0.2232†	– 0.1348†
	(0.0039)	(0.0465)	(0.0181)
Technique	0.0638†	0.1507†	0.2145†
	(0.0037)	(0.0216)	(0.0248)
Overall emissions	– 0.0015†	– 0.0725**	0.0798*
	(0.0004)	(0.0311)	(0.0415)

The market size effect refers to Eq. (25). The total effect on emissions (0.0798) refers to Eq. (29), and is correspondingly decomposed according to the sum of the market-size effect, selection effect, and upgrading effect. The scale effect refers to Eq. (30), and is correspondingly decomposed according to the sum of the market size effect, selection effect, and upgrading effect, whereas the technique effect refers to Eq. (31), and is correspondingly decomposed according to the sum of the selection effect and upgrading effect. The closed-form solutions to the numerical model results are provided in Appendix A. The standard errors are estimated using the two-step bootstrap procedure explained in the online Appendix C, wherein 1000 random samples of establishments are drawn with replacement

Significance levels: *0.10, **0.05, and †0.01

3.5.1 Numerical Model Results

Table 4 reports the numerical model results and bootstrapped standard errors, decomposed according to the effects described in Propositions 2 and 3, and Corollary 3.1.[44] The results indicate that a 1% increase in default risk increases aggregate emissions by 0.080%. Decomposing this effect indicates that a 1% increase in default risk increases the number of firms in the economy by 0.154% (market-size effect), but it also decreases average emissions by 0.074% (selection and technology upgrading effects). In turn, the selection and technology upgrading effects represent 17 and 83% (0.0015/0.0740 and 0.0725/0.0740) of the decrease in average emissions (and in turn aggregate emissions), respectively. Consequently, aggregate emissions are increasing in default risk as the increase in aggregate emissions associated with the market size effect exceeds the decrease associated with selection and technology-upgrading effects.

Dividing the effect according to scale and technique effects indicates that the increase in aggregate emissions by 0.080% is due to a decrease in emissions from the scale effect by 0.135% and an increase in emissions from the technique effect by 0.215%. In other words, the technique effect is around 60% larger than the scale effect in magnitude. Moreover, around 70% of the technique effect is due to the technology upgrading effect (0.151/0.215), while the remainder is due to the selection effect. The scale effect, on the other hand, consists of confounding effects as a 1% increase in default risk increases the number of firms in the economy by 0.154% (market-size effect), but it also decreases average output by

[44] The details of the bootstrap procedure are provided in Appendix C of the supplementary web appendix. See the table notes for the correspondence between the numerical-model coefficients and the equations in the theoretical model.

0.289%. Finally, around 77% of the decrease in average output is due to the technology-upgrading effect (0.223/0.289), while the remainder is due to the selection effect. In contrast to aggregate emissions, the magnitude of the decrease in average output exceeds the increase in the number of firms, implying that default risk decreases aggregate output.

3.5.2 Default Risk Shocks and Aggregate Pollution Emissions

This section sheds light on the economic significance of the numerical model results by investigating the repercussions of historical default risk shocks on pollution emissions and emissions intensity. I focus on the two largest shocks, the increase in Default Risk in 98/99 and the decrease in Default Risk in 02/03.

The numerical model results indicate that the 98/99 (positive) Default Risk shock would generate a 14% increase in pollution emissions and a 38% increase in emissions intensity. On the other hand, the 02/03 (negative) Default Risk shock would generate a 10% reduction in pollution emissions and a 28% reduction in emissions intensity. Therefore, historical changes in economy-wide default risk can generate economically significant long-run changes in pollution emissions and pollution emissions intensity in particular. While the magnitude of the results portend an important role of default risk in aggregate pollution emissions, it should be emphasized that the results should be interpreted as long-run effects of permanent changes in default risk, holding all other factors constant. The role of default risk shocks in short-run changes in aggregate pollution emissions is likely far more modest for a number of reasons. First, the transition from one steady state equilibrium to the new long-run equilibrium is likely to unfold over several years if not longer. Second, default risk shocks are often transitory (and subject to mean reversion) and are related to other changes in macroeconomic conditions (e.g., changes in labor force participation), which are assumed constant in the model.

3.5.3 Sensitivity Analysis

This section discusses the extent to which the numerical model results hinge on the underlying estimates and parameter values. The details of the analysis and supporting evidence are provided in Appendix E of the supplementary web appendix.

First, recall that the numerical model results are estimated using the emissions and productivity elasticities, of which the distributions of the elasticities are estimated using a two-stage bootstrap procedure wherein random samples of establishments are drawn with replacement. Figure A4 in Appendix B demonstrates that the 90% confidence interval of the distribution of the Aggregate Emissions effect (the elasticity with respect to default risk) ranges from 0.012 to 0.148, while the 90% confidence interval of the the distribution of the Aggregate Technique effect ranges from 0.14 to 0.26. Plausible ranges of the effects of default risk on aggregate emissions and emissions intensity are generally positive and economically significant, though the range of effects on aggregate emissions includes modest effects as well.

Second, I fix the emissions and productivity elasticities, and investigate the numerical model results over various ranges of parameter values.[45] Figure A6 to A9 in Appendix

[45] In particular, I assume that the elasticity of productivity and the elasticity of emissions intensity are − 0.16 and 0.89. These coefficients correspond to the baseline model as reported in Table 3, column (1), which are representative of the various model specifications.

B demonstrate that, with respect to parameter values for b and Λ, the magnitude of the Aggregate Technique and Emissions effects do not change significantly over a wide range of parameter values, and the magnitude of the effects are generally lower bounds of potentially larger effects. Similarly, with respect to parameter values δ and α, the magnitude of the Aggregate Technique and Emissions effects do not change significantly over a wide range of parameter values, though it's not possible to bound the effects. In general, the magnitude of the Aggregate Technique effect exhibits some variation over the range of parameter values, but the effect remains economically significant in all instances (> 0.17). The magnitude of the Aggregate Emissions effect also exhibits some variation over the range of parameter values, but the effect remains strictly larger than zero in all instances (> 0.03). Thus, while the numerical model results exhibit some degree of sensitivity to the parameter values chosen, the qualitative conclusions remain unchanged.

4 Conclusion

This paper employs a general equilibrium model to analyze the effect of default risk on aggregate pollution emissions. The model demonstrates that, by raising borrowing costs, default risk reduces investment in technology upgrading, and results in an economy consisting of more numerous firms, but firms that are on average less productive and more pollution intensive. Specifically, this paper demonstrates that default risk generates three conceptually distinct effects, which I refer to as the (1) market-size effect, (2) technology-upgrading effect, and (3) selection effect.

The empirical analysis explores the effect of default risk on pollution emissions and productivity for a panel of manufacturing facilities. Using a two-stage procedure where default risk is estimated in the first stage, the results indicate that the estimated elasticity of emissions intensity and productivity with respect to default risk is 0.89 and -0.16, respectively. This paper also employs an alternative difference-in-differences approach, exploiting variation in the magnitude of default risk shocks across industries, which yields consistent results.

This paper uses the coefficients from the empirical analysis, coupled with parameters estimated by previous studies, to estimate the effect of an increase in the economy-wide default risk on aggregate pollution emissions, and to divide the effects into market-size, technology-upgrading, and selection effects. The numerical model results demonstrate that average pollution emissions are decreasing in default risk, with most of the effect due to the technology-upgrading effect, while aggregate pollution emissions are increasing in default risk due to the confounding market-size effect. Moreover, because aggregate output is decreasing in default risk, emissions intensity is increasing in default risk to an even greater extent than aggregate emissions.

What insights does the paper confer in understanding the environmental implications of changes in the macroeconomy, including marked changes such as the 2008 Great Recession? According to the model, the increase in default risk associated with the Great Recession would have resulted in a 3.2% increase in aggregate emissions, and a 8.6% increase in aggregate emissions intensity (holding all other factors constant).[46] There are several

[46] Consistent with this paper, Leibovici and Wiczer (2019) document that default risk increased during the Great Recession, as evidenced by changes in credit ratings and exit behavior.

reasons to expect that observed changes might be at variance with the model predictions. As mentioned, the model analyzes long-run changes associated with permanent changes in default risk, rather than cyclical changes in the macroeconomy. Moreover, the model isolates the effect of changes in default risk holding all other factors constant, whereas macroeconomic shocks, such as the Great Recession, are associated with numerous changes that might bear on pollution emissions, such as countercyclical government policies and transitory changes in the labor force. For instance, the model predicts a unit elasticity of aggregate emissions with respect to aggregate labor supply, implying that the reduction in emissions associated with the reduction in employment during the Great Recession (around 7%) would be larger in magnitude than the increase in emissions associated with the increase in default risk. Despite the aforementioned qualifications and caveats, the predictions of the model are mostly consistent with the limited existing empirical evidence, at least qualitatively. For example, based on satellite measurements, Tong et al. (2016) estimate that the Great Recession reduced nationwide surface ozone air pollution by around 2–3% (relative to a business-as-usual scenario). After accounting for the reduction in emissions associated with the reduction in employment, the model predicts that emissions would decline by around 3–4%, which is comparable to the observed changes.

This paper represents a first step in understanding the role of default risk in aggregate pollution emissions; however, there are many avenues for future research. For example, while the results of this paper are relevant to a composite measure of pollution emissions for the manufacturing sector, future research might investigate alternative pollution emissions, such as carbon dioxide emissions, and explore the effects across industries within the manufacturing sector or other industries, such as emissions generated by the electricity-generation sector. Future research might also attempt to identify other sources of exogenous variation in default risk, to rule out potential bias. Finally, this paper might be more intricately linked to the broader literature investigating the role of macroeconomic dynamics in pollution emissions.

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